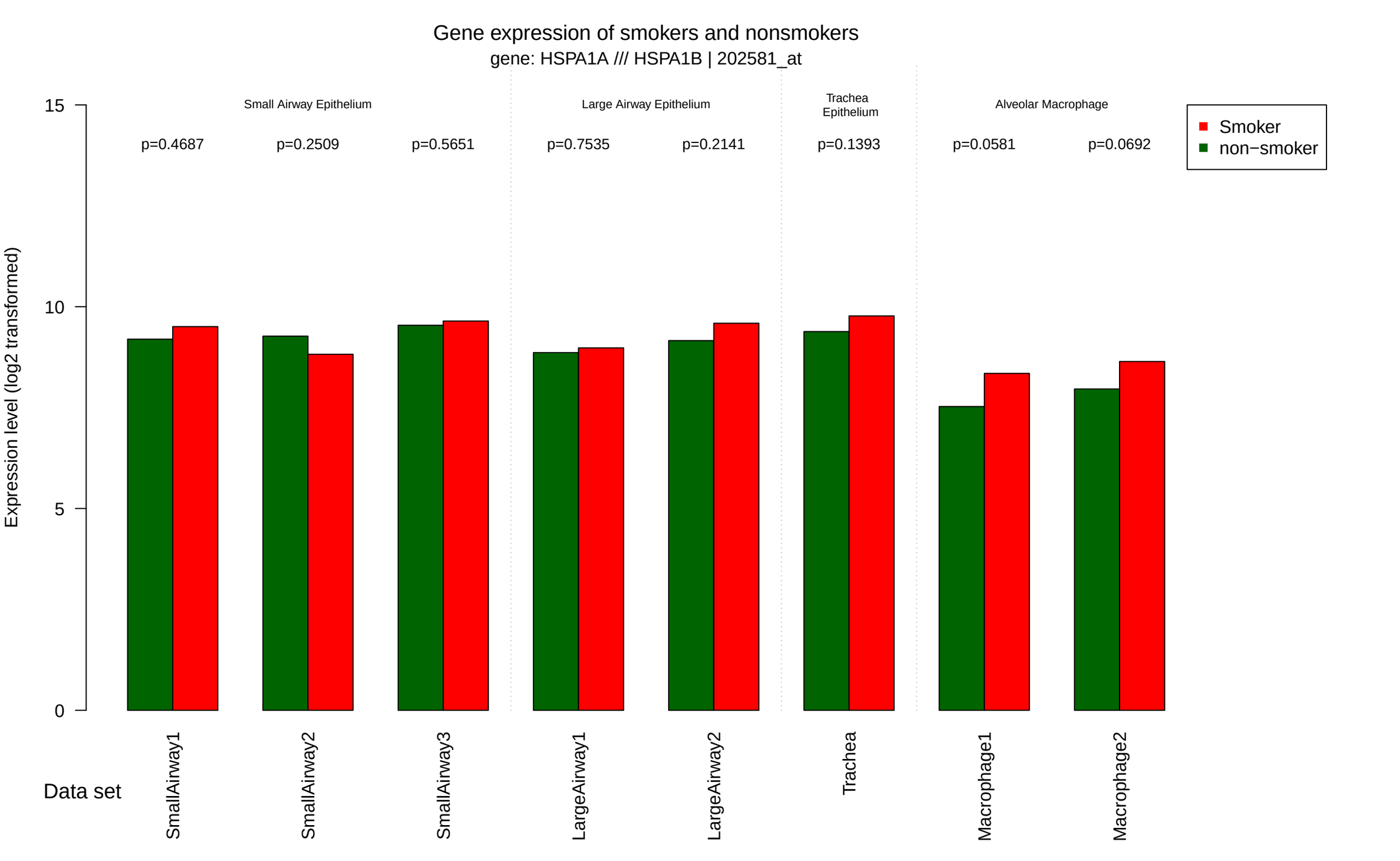
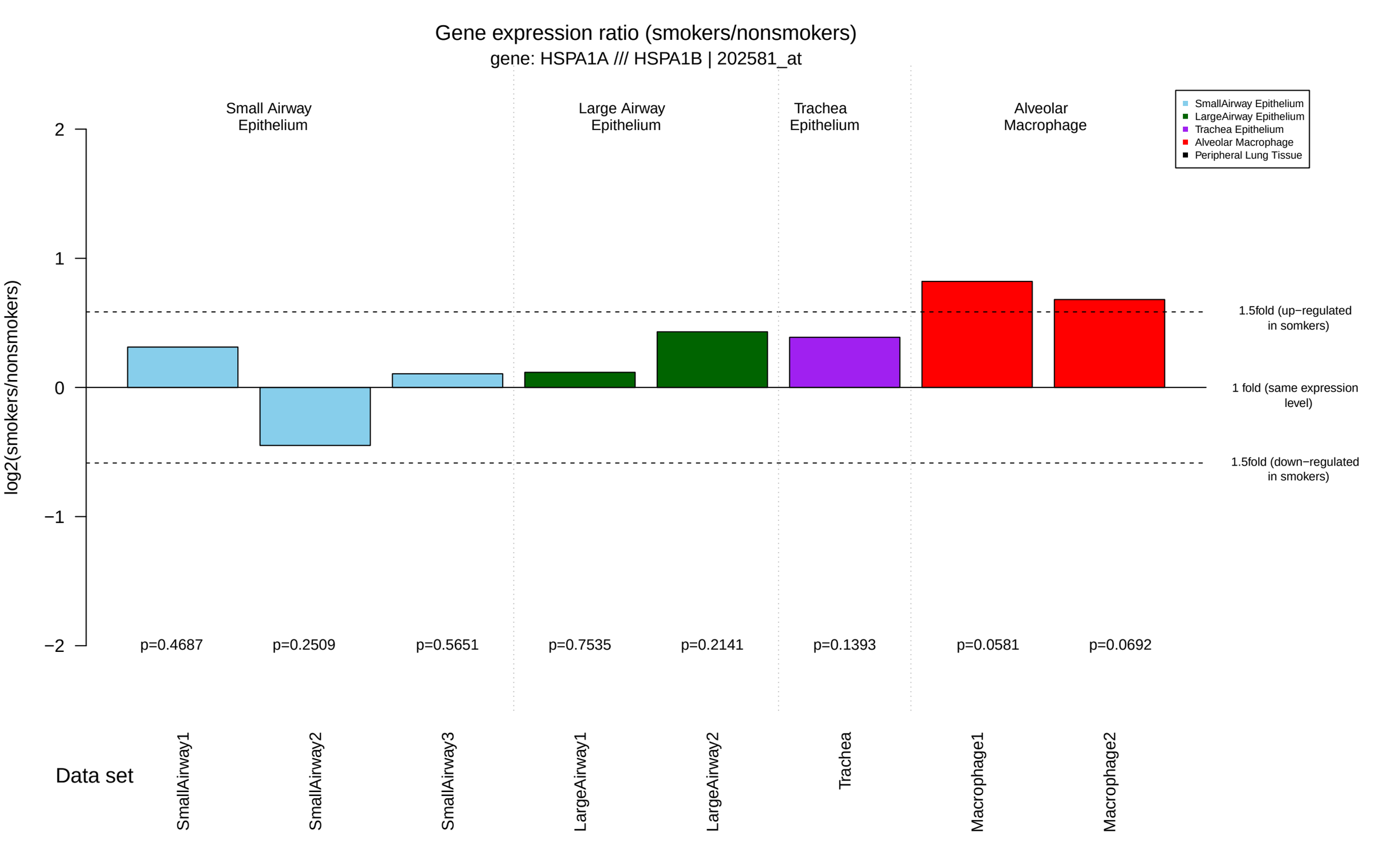
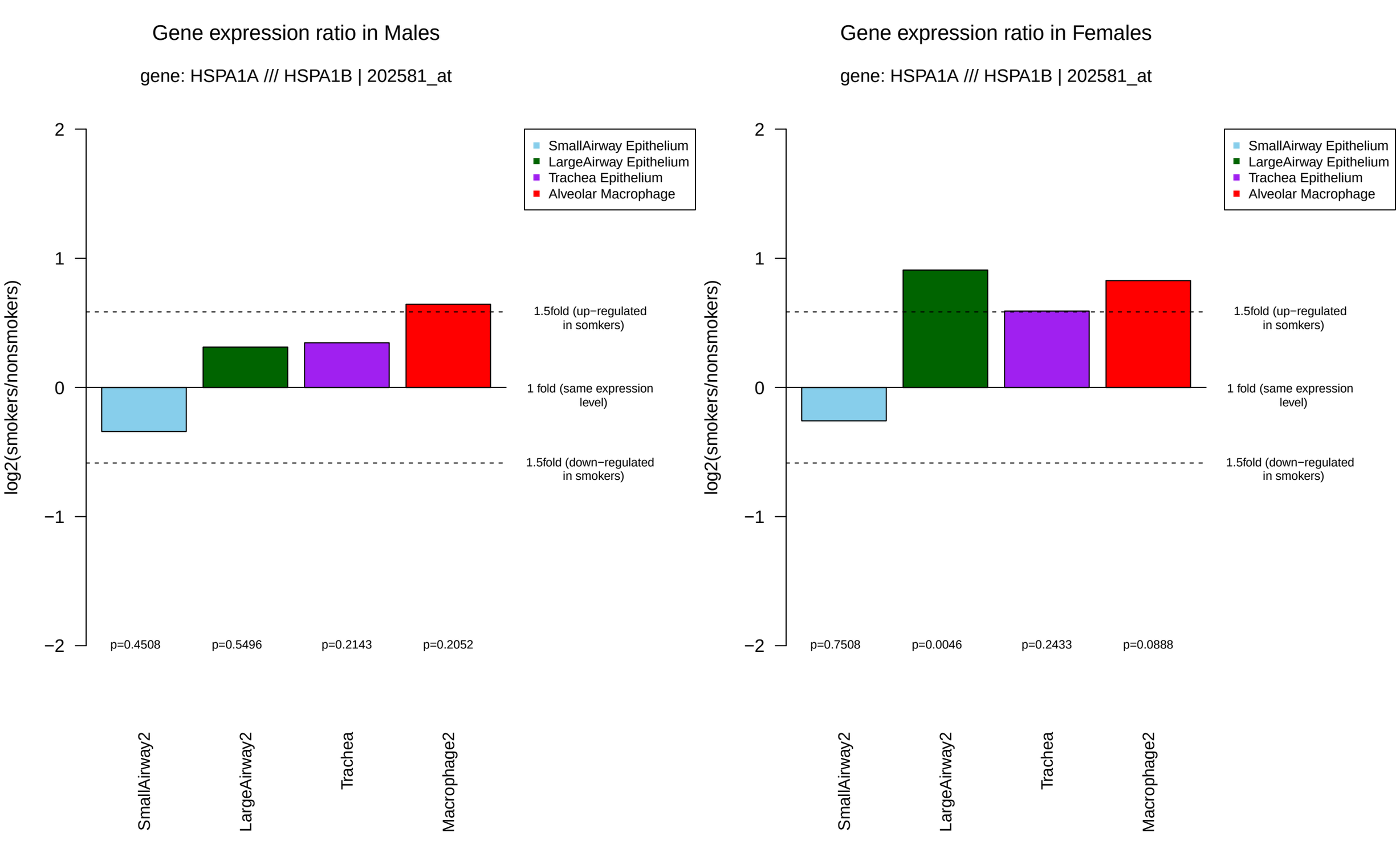
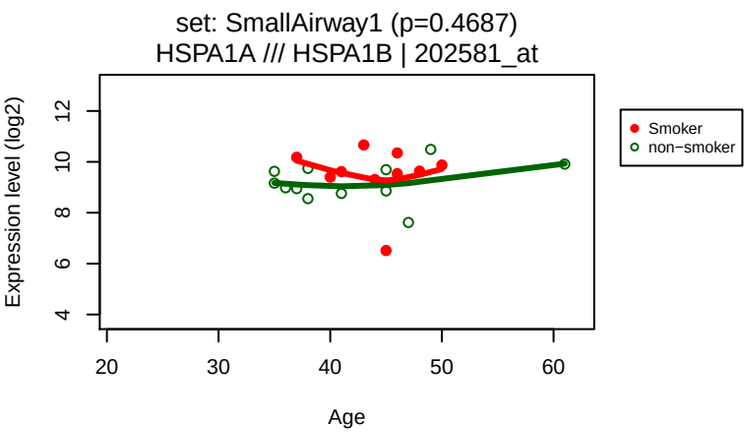
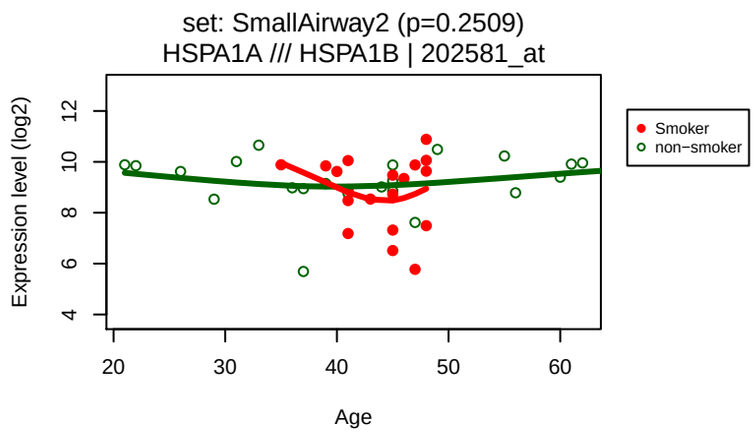
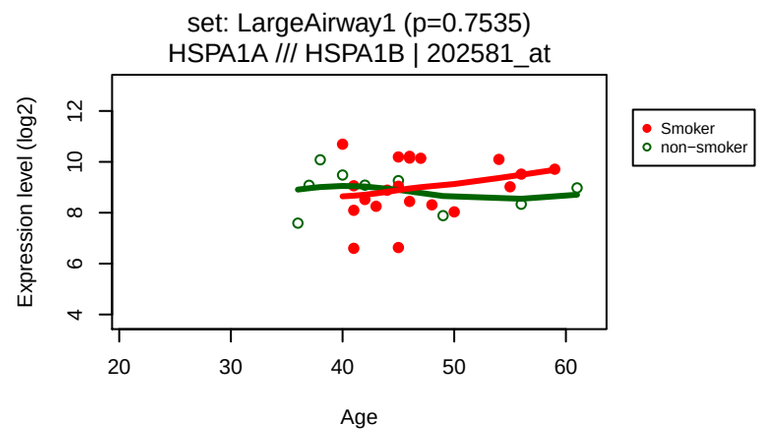

Gene expression of smokers and nonsmokers
gene: HSPA1A /// HSPA1B | 202581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4687
p=0.2509
p=0.5651
p=0.7535
p=0.2141
p=0.1393
p=0.0581
p=0.0692
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

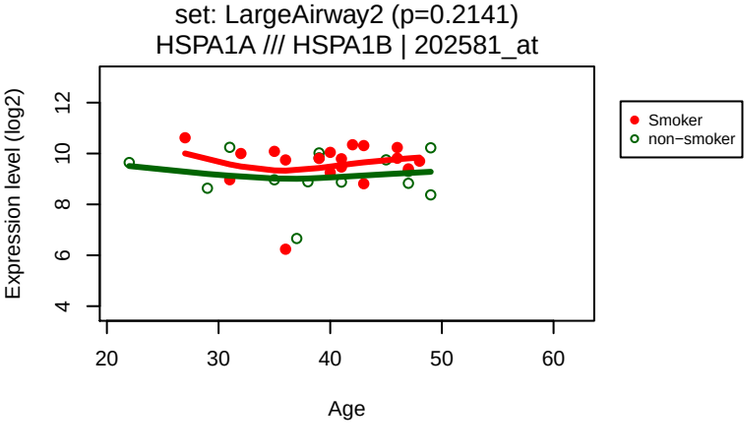

Gene expression ratio (smokers/nonsmokers)
gene: HSPA1A /// HSPA1B | 202581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4687
p=0.2509
p=0.5651
p=0.7535
p=0.2141
p=0.1393
p=0.0581
p=0.0692
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

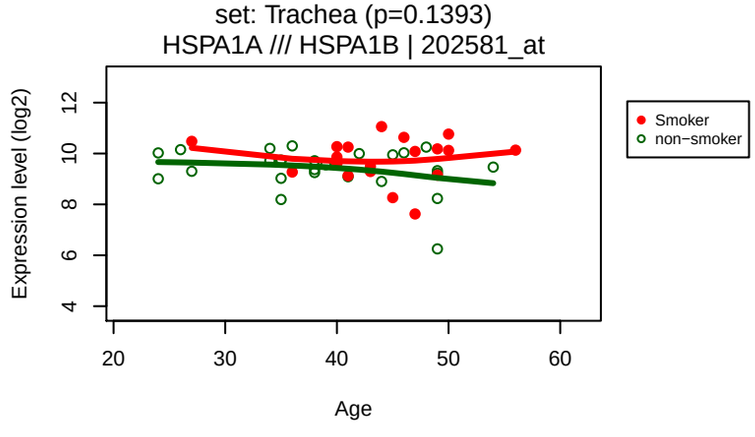

Gene expression ratio in Males
gene: HSPA1A /// HSPA1B | 202581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4508
p=0.5496
p=0.2143
p=0.2052
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA1A /// HSPA1B | 202581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7508
p=0.0046
p=0.2433
p=0.0888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

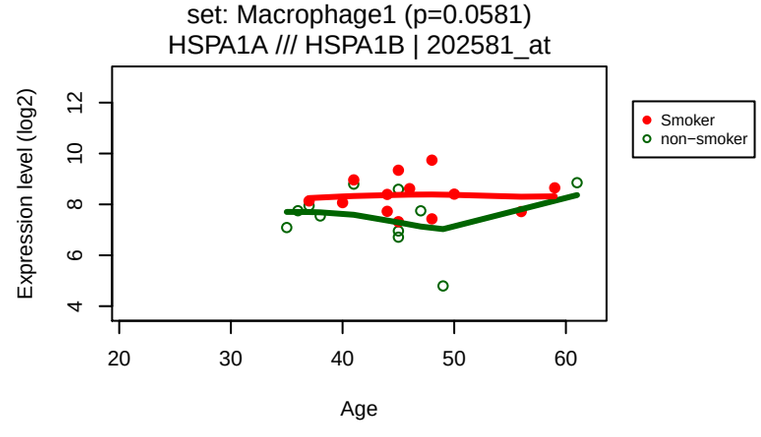

set: SmallAirway1 (p=0.4687)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker

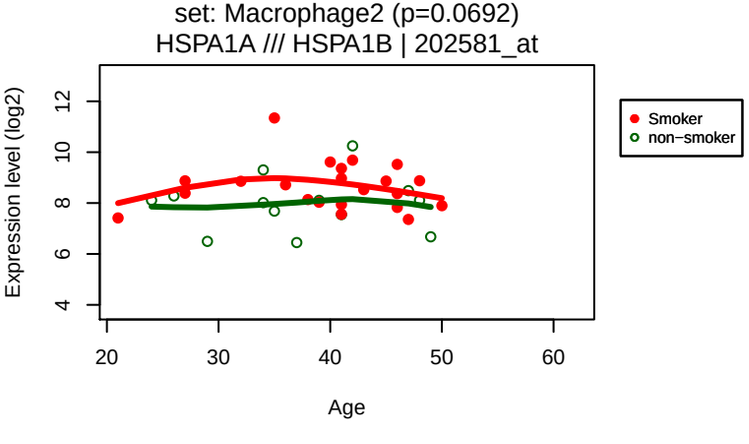

set: SmallAirway2 (p=0.2509)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker

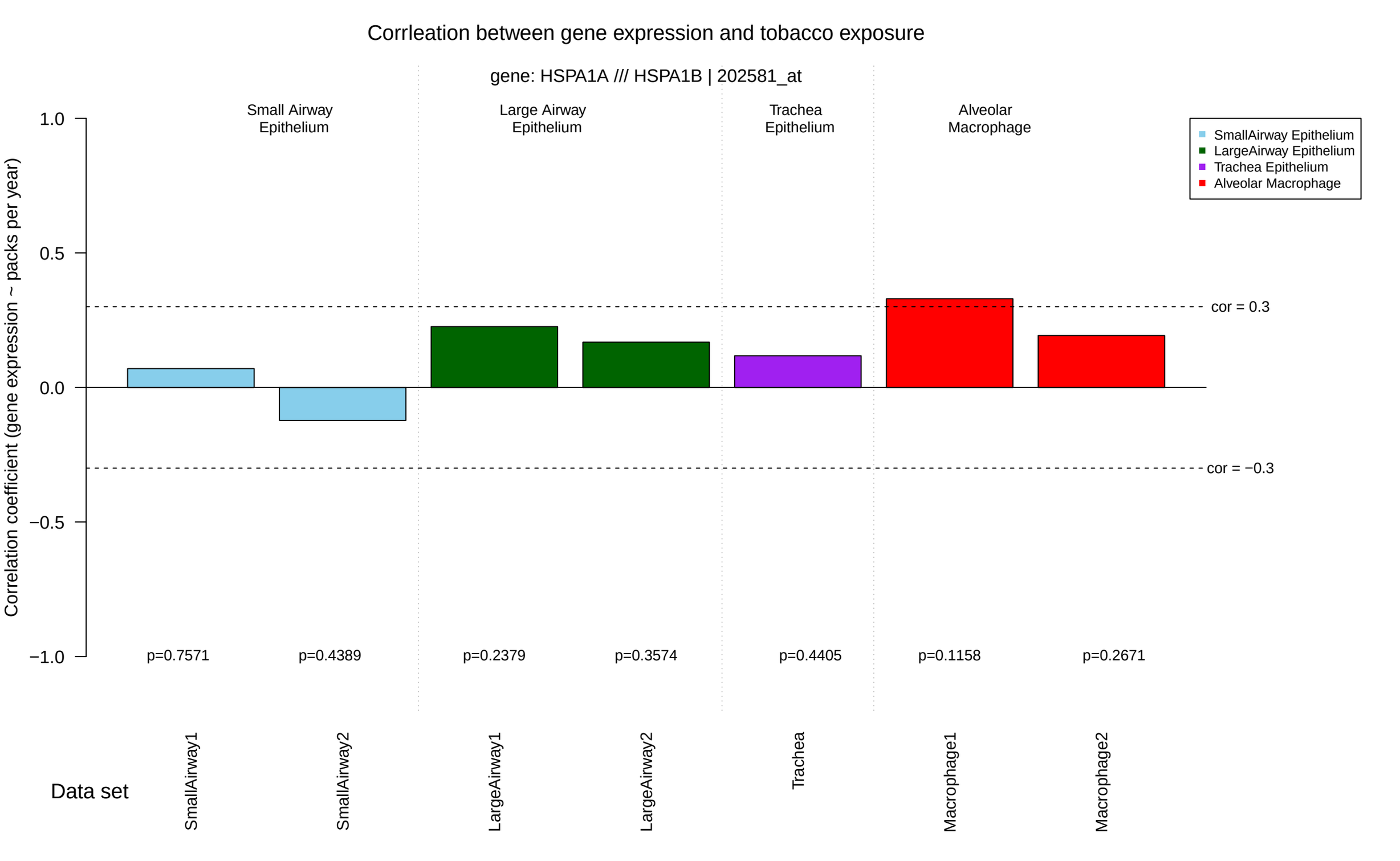

set: LargeAirway1 (p=0.7535)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2141)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1393)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0581)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0692)
HSPA1A /// HSPA1B | 202581_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA1A /// HSPA1B | 202581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7571
p=0.4389
p=0.2379
p=0.3574
p=0.4405
p=0.1158
p=0.2671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2